A crossing that exchanges the remaining two rows

Human Compression project: L044 continuation

Lemma 1 (A pure alternating group from two rows)**.** *Let $t \geq 3$ and let the distinguished points be $b, a_0, a'_0, \ldots, a_{t-1}, a'_{t-1}$. Suppose*

$$q_i = (a_{i-1}\ a_i)(a'_{i-1}\ a'_i) \quad (1 \leq i < t), \qquad s = (b\ a_0)\prod_{i=1}^{t-1}(a_i\ a'_i)f,$$

where f is an involution supported outside these $2t+1$ points. Then $\langle s, q_1, \ldots, q_{t-1}\rangle$ contains the pure natural alternating group on all $2t+1$ distinguished points.

Proof. The permutation $c = (sq_1)^2$ is a pure five-cycle on $T = \{b, a_0, a'_0, a_1, a'_1\}$: on T, the product sq_1 is a five-cycle, and off T all factors have order at most two and disjoint supports. Furthermore q_1 is a reflection of that five-cycle and fixes b. Thus $D = \langle c, q_1\rangle$ is the natural dihedral group of order ten on T.

Choose its reflection d fixing a_1. It moves a'_1 to a different point z of $T \setminus \{a_1\}$. Since d fixes a_2, a'_2, the product $q_2 q_2^d$ is a pure three-cycle on $\{a'_1, a'_2, z\}$: the common transposition $(a_1\ a_2)$ cancels. Conjugating this cycle by powers of c gives three-cycles on $\{a'_2, u, v\}$, where the pairs $\{u, v\}$ run through a connected five-cycle on T. Indeed every nonzero step in the cyclic group of order five generates it.

For any connected graph on T, the three-cycles through a fixed additional point and its edges generate the natural alternating group on that enlarged set. To see this, take a spanning tree and add its edges from a starting edge: each new three-point alternating group overlaps the preceding support in the additional point and one tree vertex, so the elementary alternating-overlap lemma grows the entire alternating group. We therefore obtain pure $\mathrm{Alt}(T \cup \{a'_2\})$. Its conjugate by q_2 has a different six-point support with five points in common. Their union is $T \cup \{a_2, a'_2\}$, so overlap gives its pure A_7.

Successively conjugate by $q_3, \ldots, q_{t-1}$. At each step exactly the next pair enters the union, while all but two points of the old support remain in the intersection. These intersections have size at least five. Repeated overlap gives the claimed pure alternating group. The external factor f never enters this group: it was eliminated already in $(sq_1)^2$. □

Theorem 2 (Exclusion of the row-swap crossing)**.** *Retain the complete critical-head hypotheses, actual IP and all intransitive ambient omissions. Suppose a later unique interior cut has adjacent first shared label g and crossing $j = g+1$. Write its omission orbits as $C = C_g$ and A, with crossing $(b\ a)$, $b \in C$. Suppose the reversed support equality has*

$$|A| = 2t, \quad t = r - g - 1, \quad \mathrm{Fix}(P_{g-1}) = A \cup \{b\}, \qquad P_k = G_{\leq k},$$

and $Q = G_{>j}$ has two faithful natural S_t orbits on A. Let $v = \rho_g|_A$ exchange these two orbits equivariantly. For $t \geq 4$, the crossing $s = \rho_j$ fixes every point of A other than a. More precisely, it exchanges a, b and fixes $a' = a^v$ and every point of $A \setminus \{a, a'\}$. The alternative in which it swaps the two remaining rows is impossible, also when this alternative is specified for $t = 3$.

Proof. Use v to identify the two rows, and choose one point of each pair in the first row. The pair action of Q is S_t. Its distinguished pair omissions are intransitive: if one were transitive, adjoining ρ_g would join the two rows inside the corresponding ambient omission, making A one orbit there.

Since the omitted mark is supported on A, it could join no ambient orbits, contradicting full transitivity. A selected edge per pair-image label is a forest: a cycle would give an avoiding path for one selected omission crossing. Its $t-1$ edges on t vertices make it a tree. Deleting an edge leaves exactly the two orbits of that omission, since the remaining tree edges connect both sides. Each image preserves all the other omission cuts; their common classes are singletons and the endpoints of its own edge. Thus each image is precisely that transposition. Nonadjacent labels have disjoint edges by commutation, so no vertex has degree three and the tree is a path. The unique-crossing anchor is fixed by all but the first Q generator. Thus the actual Q marking is

$$q_i = (a_{i-1}\ a_i)(a'_{i-1}\ a'_i), \quad a_0 = a, \quad a'_0 = a', \qquad 1 \le i < t.$$

These facts concern the actual faithful row actions; no projected intersection property is used.

The subgroup $Q_1 = \langle q_2, \dots, q_{t-1} \rangle$ has fixed set exactly $C \cup \{a, a'\}$ when $t \ge 3$. Since s centralizes Q_1, the unique crossing forces it to fix a' and preserve the remaining paired points. When $t \ge 4$, the two nontrivial Q_1-orbits are natural S_{t-1} orbits of size at least three. Their individual centralizers are trivial, and their equivariant identification is unique. Thus the restriction of s to their union is either the identity or the full row swap.

Suppose it is the row swap, for any $t \ge 3$. Then

$$s = (b\ a) \prod_{i=1}^{t-1} (a_i\ a'_i) f, \qquad \operatorname{supp}(f) \subseteq C \setminus \{b\}.$$

Lemma 1 gives pure $\mathrm{Alt}(A \cup \{b\}) \le G_{\ge g+1}$. The generator ρ_g moves b, because P_{g-1} fixes b and P_g is transitive on C. Set $c = b^{\rho_g} \in C \setminus \{b\}$. It preserves A, so conjugation by ρ_g and overlap give

$$\mathrm{Alt}(A \cup \{b, c\}) \le G_{\ge g}.$$

On the prefix side, P_g has exterior image $\langle v \rangle$. The crossing attaches a and then v attaches a'; both generators preserve the other paired points. Hence $C_{g+1} = C \cup \{a, a'\}$, and the complete prefix theorem puts its pure alternating group in P_{g+1}. Actual IP would therefore put pure $\mathrm{Alt}(\{a, a', b, c\})$ in

$$P_{g+1} \cap G_{\ge g} = \langle \rho_g, s \rangle.$$

This is impossible: a group generated by two involutions is dihedral, with all odd-order elements in its cyclic rotation subgroup, whereas A_4 has noncommuting three-cycles. Only the identity restriction on the remaining rows survives when $t \ge 4$. □

Attribution and remaining scope. The source route is L044 Propositions 5.20–5.32. The acquired pure-prefix and alternating-overlap proofs remain paid. The five-cycle/reflection construction supplies an explicit new pure alternating subgroup, and the final actual-IP obstruction reuses the acquired noncommuting-three-cycle argument. The identity restriction for $t \ge 4$, the additional small centralizer possibilities at $t = 2, 3$, other cuts, opposite heads and terminal singleton remain unresolved here. No new computation, formal theorem, whole-original or question credit is claimed.